

**DATA AND ETHICS
FINAL REPORT**

**Bias in Artificial Intelligence Systems:
Who Is Responsible The Company, the Programmer, or the Data?**

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Abstract

Artificial intelligence systems now make or inform high stakes decisions in hiring, credit assessment, healthcare allocation, and criminal justice. When these systems perpetuate or amplify historical bias, real harm results yet accountability remains ambiguous across three candidate parties: the companies that deploy AI, the programmers who build it, and the data sources that encode historical discrimination. This survey paper synthesizes existing empirical and theoretical literature to examine the distribution of responsibility for AI bias. Drawing on documented cases such as commercial facial recognition disparities and healthcare allocation algorithms and situating the analysis within both European regulatory frameworks and Saudi Arabia's emerging AI governance structures under SDAIA, the paper argues that deploying companies bear primary ethical and legal responsibility for algorithmic harm. It acknowledges the competing arguments for individual programmer accountability and diffuse societal causation and concludes by identifying gaps in current regulatory frameworks that must be addressed for accountability to be meaningful.

Keywords: AI bias, algorithmic accountability, machine learning fairness, corporate responsibility, SDAIA, regulatory frameworks, Saudi Arabia

Introduction

Artificial intelligence systems have become embedded in decisions that used to require human judgment. Automated hiring platforms screen thousands of resumes before any recruiter reviews them. Credit scoring algorithms determine loan eligibility in milliseconds based on patterns extracted from historical financial data. Healthcare AI systems triage patients and allocate resources. In some jurisdictions, predictive tools directly influence pretrial detention and sentencing recommendations in criminal courts. These are not experimental or peripheral applications, they are operational systems affecting real people every day, often at scale and often without meaningful human review of individual decisions.

The core ethical problem is that the data these systems learn from is not neutral. Machine learning models identify patterns in training data; they do not construct independent judgment. When training data reflects decades of discriminatory hiring practices, racially unequal policing, or gender biased assumptions embedded in historical medical research, the model learns to replicate those patterns and then scales and automates them. Discrimination does not require intent; it is inherited from the historical record and reproduced systematically.

Several cases have made this structural problem concrete and well documented. Buolamwini and Gebre (2018) found that commercial facial recognition systems misclassified darker skinned women at error rates more than thirty percentage points higher than lighter skinned men, revealing intersectional bias in products already deployed commercially. Obermeyer et al. (2019) demonstrated that a widely used healthcare algorithm underestimated the medical needs of Black patients relative to white patients with equivalent clinical conditions, because the model used historical healthcare spending as a proxy for health needs a proxy that encoded preexisting racial disparities in access to care. In both cases, the bias was invisible to users and only surfaced through careful external auditing.

What makes AI bias particularly difficult as a governance problem is not simply that bias exists bias is present in many human run processes, but that AI embeds it at scale, in opaque models, with outcomes that are difficult to contest. When a human decision maker makes a biased determination, there is at least one person who can be identified, questioned, overruled, or disciplined. When an algorithm makes the same determination for a hundred thousand applicants simultaneously, and the decision logic is locked inside a proprietary model, accountability becomes genuinely ambiguous.

This paper investigates that ambiguity. Its central question is: To what extent should companies be held primarily responsible for bias in AI systems compared to programmers and data sources, and what regulatory frameworks can most effectively enforce accountability? The paper proceeds as follows:

1. surveys the empirical literature on AI bias.
2. analyzes the three candidate loci of responsibility.
3. examines existing and proposed regulatory frameworks, with particular attention to the Saudi context.
4. presents the paper's central argument.
5. addresses counterarguments.
6. identifies research gaps and conclusions.

1. Survey of the Literature on AI Bias

1.1 Defining and Classifying Bias in Machine Learning

Mehrabi et al. (2021) provide the most comprehensive taxonomy currently available, identifying and categorizing over twenty distinct types of bias in machine learning systems. They distinguish among historical bias (bias present in the social world that is reflected in data collection), representation bias (systematic underrepresentation of subgroups in training data), measurement bias (systematic error introduced through the choice or operationalization of proxy variables), aggregation bias (treating heterogeneous groups as homogeneous during modeling), and evaluation bias (benchmarking on datasets that do not represent the deployment population). Their review makes clear that bias in AI systems is not a single, uniform phenomenon it is a family of related failures with different origins, different mechanisms, and different remediation requirements.

This complexity has practical implications for accountability. If the bias originates in historical data structures (historical bias), responsibility may plausibly be attributed to systemic social forces. If it originates in decisions about variable selection or proxy use (measurement bias), responsibility is more clearly traceable to technical choices made by specific individuals. If it originates in how a product is tested and validated before deployment (evaluation bias), responsibility falls more directly on the organization that controls the deployment process.

1.2 Empirical Evidence of Bias in Deployed Systems

The empirical literature on AI bias has grown substantially since 2016. Buolamwini and Gebru's (2018) Gender Shades audit established that commercial facial recognition systems from three major technology companies produced significantly higher error rates for darker skinned faces and for female

faces, with the intersection of both characteristics producing the worst performance. This finding was particularly significant because the companies involved had marketed these products as generally reliable, and their internal benchmarking had not revealed the disparate performance the disparity was only identified through an external audit using a more representative benchmark dataset.

Obermeyer et al. (2019) examined a healthcare risk stratification algorithm used across hundreds of health systems in the United States to identify patients for complex care management programs. They found that the algorithm systematically assigned lower risk scores to Black patients than to white patients with the same severity of illness. The root cause was a proxy variable problem: the model used healthcare spending as a proxy for health need, but because Black patients had historically spent less on healthcare due to unequal access, not lower need the model learned to rank them as healthier. This case is notable because the algorithm was working exactly as designed; the failure was in the choice of the proxy variable and the failure to audit for disparate impact before deployment.

Lambrecht and Tucker (2019) studied algorithmic bias in a different domain, the display of STEM career advertisements on social media platforms. They found that even when advertisers explicitly targeted both genders, platform optimization algorithms delivered ads disproportionately to men, because women were a more expensive audience to reach in general advertising markets. The bias emerged not from any discriminatory intent but from optimization dynamics within the advertising auction system. This case illustrates how bias can emerge as an emergent property of system design rather than from biased training data in the conventional sense.

1.3 Accountability and Auditing Frameworks

Raji et al. (2020) propose an end to end framework for internal algorithmic auditing that covers the full lifecycle from problem formulation through deployment and monitoring. Their framework emphasizes that meaningful accountability requires audit processes that are integrated into development from the beginning not applied as a retrospective check before launch. They identify the lack of organizational mechanisms for accountability as a central failure mode, noting that many companies have the technical capacity to audit their systems but lack the governance structures that would require them to do so systematically. Critically, their framework distinguishes between technical auditing testing model performance across demographic groups and organizational auditing, which examines the processes, incentives, and governance structures that determine whether technical findings are acted upon.

Diakopoulos (2016) examines accountability in algorithmic decision making from a journalistic and public interest perspective, arguing that opacity in algorithmic systems fundamentally undermines the public's ability to hold decision makers accountable. He identifies four key aspects of algorithmic systems that require scrutiny: the inputs they use, the logic by which they transform inputs into outputs, the outputs they produce, and the feedback loops through which outputs affect future inputs. His analysis highlights the structural conflict between corporate interests in protecting proprietary models and the public interest in understanding consequential automated decisions.

1.4 Ethical Frameworks and Global AI Governance

Jobin, Ienca, and Vayena (2019) conducted a systematic review of eighty four AI ethics guidelines published by governments, corporations, professional bodies, and civil society organizations worldwide. They found broad convergence on five principles: transparency, justice and fairness, non maleficence, responsibility, and privacy. However, they also found significant divergence in how these principles are defined and operationalized across different cultural and institutional contexts, and they note that the proliferation of principles-based frameworks has not been matched by effective enforcement mechanisms. This finding has important practical implications: the existence of an ethics guideline does not, by itself, produce accountable behavior.

Floridi et al. (2018), writing within the AI4People initiative, develop a more structured ethical framework grounded in four classical bioethical principles beneficence, non-maleficence, autonomy, and justice and extend them to the AI context. They argue explicitly that individual professionals in high stakes technical roles carry a duty of care that is not fully transferable to their employer, paralleling the established ethical frameworks of medicine and engineering. They also introduce a fifth principal explicability as essential to AI ethics specifically. A system that cannot be explained cannot be effectively audited, and a system that cannot be audited cannot be effectively held accountable.

1.5 The Saudi Arabian Context

Alelyani and Kumar (2021) conducted a systematic literature review of AI adoption in Saudi Arabia, covering research published through 2020. They found that the existing literature focused primarily on technical performance and economic adoption metrics, with substantially less attention to governance, ethical risk, and accountability structures. This gap is significant given the scale of AI integration underway in the Kingdom as part of Vision 2030. The healthcare AI market in Saudi Arabia reached \$211 million in 2024 and is projected to reach \$805 million by 2030. SDAIA's published ethics principles and the Personal Data Protection Law (PDPL, Royal Decree M/147) establish general

frameworks invoking fairness, transparency, and human oversight, but they have not yet specified in operational terms how liability for algorithmic bias should be assigned when harm occurs to individuals.

2. The Three Candidates for Responsibility

2.1 The Deploying Company

The case for primary corporate responsibility rests on several converging grounds. First, companies make the deployment decision. They determine whether to purchase or build an AI system, which populations apply it to, and what performance thresholds must be met before a system goes to market. They control the conditions under which the system operates. Second, companies profit from the system's output. When an AI hiring tool reduces human review time, the company captures that efficiency gain. Third, companies have the resources to commission audits, engage independent testing organizations, monitor system performance post deployment, and respond to identified harms. The claim that a well resourced corporation bears no responsibility for the known failure modes of a system it profits from is difficult to defend on any reasonable account of ethical responsibility.

Wachter, Mittelstadt, and Russell (2021) make an additional structural argument: because fairness has multiple mathematically incompatible definitions, no purely technical solution can fully resolve AI bias. The decision about which definition of fairness to apply and therefore which group's interests to prioritize when trade offs arise is inherently a normative and social decision, not a technical one. Companies, as the entities that make deployment decisions, are the natural locus of that normative accountability.

2.2 The Individual Programmer

A competing view holds that programmers and data scientists who build these systems carry significant individual moral responsibility that cannot simply be absorbed by their employer. Mittelstadt, Russell, and Wachter (2019) argue that technical expertise creates specific obligations: professionals who understand the limitations of a training dataset, the failure modes of a model architecture, or the disparate impact of a proxy variable have a duty to document those limitations, communicate them clearly, and resist deployment when known risks are serious and unaddressed. This parallels the established ethics of engineering and medicine, where individual professionals are held to standards independent of their institutional employer.

Under Saudi law, this argument has legal as well as ethical weight. The PDPL and the Anti Cybercrime Law (Royal Decree M/17) both create provisions for individual accountability when gross professional negligence or deliberate misconduct can be established. SDAIA's ethics principles similarly emphasize individual human oversight as a required safeguard for sensitive automated decisions.

2.3 The Data and Its Historical Origins

The deepest challenge to assigning responsibility to any specific actor is the argument from historical causation. Mehrabi et al. (2021) emphasize that historical bias rooted in long standing patterns of discrimination that predate AI entirely is among the most difficult categories to address precisely because no single contemporary actor created it. Racial disparities in healthcare access, gender discrimination in hiring, and racially unequal criminal justice practices developed over decades or centuries. A company that trains a model on historical hiring decisions is, in a sense, automating discrimination it did not itself originate.

The historical causation argument is at its strongest when the data bias is genuinely unavoidable when no plausible alternative dataset exists and when the use of historical data is necessary to achieve a legitimate and beneficial purpose. It is at its weakest when organizations had known alternatives they chose not to use, when they failed to audit for disparate impact before deployment, or when they continued operating a system after its biased outputs were identified. The proxy variable problem at the center of the Obermeyer et al. (2019) case was identifiable before deployment; the algorithm's developers chose not to test for it. The historical causation argument provides a partial explanation for why bias exists in the data; it does not provide a full justification for deploying a system without auditing for its impact.

2.4 Toward a Shared but Hierarchical Responsibility Model

The three candidates for responsibility are not mutually exclusive; they operate at different levels of analysis and can coexist within a comprehensive accountability framework. At the organizational level, the deploying company is the primary accountable entity. It controls the deployment decision, benefits from the system's outputs, and has the greatest capacity to prevent harm through systematic pre deployment auditing and ongoing monitoring.

At the individual level, programmers and data scientists carry secondary but genuine accountability grounded in professional ethics. Their obligation is not to ensure that the system produces no bias that may often be impossible but to document known limitations honestly, communicate them to decision makers, and resist deployment when known risks are serious and unaddressed. At the societal level,

historical and structural bias requires policy interventions beyond AI regulation specifically including anti-discrimination law, diversity in data collection programs, and investments in producing more representative training datasets.

3. Regulatory Frameworks

3.1 European Union Approaches

The EU AI Act and GDPR's automated decision making provisions (Articles 13 15, 22) represent the most developed mandatory regulatory framework currently in effect. They require transparency, provide rights to explanation and human review for consequential automated decisions, and classify AI systems by risk level with high-risk applications in employment, credit, and healthcare subject to conformity assessment before deployment. Wachter, Mittelstadt, and Russell (2021) analyze the relationship between these frameworks and AI fairness requirements in depth, finding significant gaps that must be addressed to make legal accountability effective.

A central finding of Wachter et al.'s analysis is that the GDPR's non-discrimination provisions and the multiple mathematical definitions of algorithmic fairness are fundamentally incompatible in many cases. Because satisfying one definition of fairness frequently requires violating another, there is no universal technical solution that satisfies all fairness criteria simultaneously. This means that compliance with non-discrimination law in the context of AI requires normative choices that cannot be delegated to technical systems and must be made by accountable human and organizational actors which is precisely why organizational accountability is the appropriate primary locus for regulatory requirements. The EU AI Act's risk classification approach offers a practical model for other jurisdictions, including Saudi Arabia, to adapt.

3.2 Saudi Arabian Governance Structures

SDAIA's AI Ethics Principles identify fairness, transparency, human oversight, and accountability as core governance values. The PDPL establishes obligations for data controllers to determine the purpose and means of data processing covering collection, use, and automated decision making. Entities that process personal data through biased algorithms face liability for discriminatory outcomes under the PDPL framework. The Anti Cybercrime Law addresses the criminal dimension when deliberate manipulation of data or systems can be established.

The current Saudi framework creates a principled foundation but lacks the operational specificity needed for effective enforcement. Key gaps include the absence of mandatory pre deployment auditing

requirements for high risk AI applications, the absence of designated enforcement authorities with technical capacity to investigate algorithmic harm, and the absence of individual right to explanation provisions for automated decisions comparable to those in the GDPR. Closing these gaps is necessary for the framework to translate from principles to effective accountability.

4. Central Argument: Primary Corporate Responsibility

Synthesizing the empirical literature and regulatory analysis, this paper argues that deploying companies should bear primary responsibility for AI bias, while acknowledging that individual programmers and historical data structures contribute to the problem and must also be addressed within a complete accountability framework.

The case for primacy of corporate responsibility rests on four grounds. First, the deployment decision is the critical intervention point. Whatever biases exist in training data or in specific technical choices, it is the company that decides to put the system into consequential use. That decision is the proximate cause of any harm the system causes. Second, companies have the greatest capacity to prevent harm through auditing requirements, diverse development teams, independent testing, and post deployment monitoring. The argument that capacity entails responsibility is well established in both ethical theory and legal doctrine. Third, companies benefit from the system's output; it is a fundamental principle of accountability that those who profit from an activity bear responsibility for its costs. Fourth, as Raji et al. (2020) argue, the accountability gap in AI is fundamentally an organizational problem the absence of governance structures that require companies to audit their systems systematically, not primarily a technical one.

This does not mean programmers bear no responsibility. Floridi et al. (2018) are correct that professional expertise creates ethical obligations that are not fully absorbed by employment. But individual professional accountability and organizational accountability are not zero sum. A framework that holds both companies and individual practitioners accountable as the medical profession does through both institutional and individual licensing and liability structures is more robust than one that attributes responsibility exclusively to either.

It is also worth noting that the argument for primary corporate responsibility is supported by what has actually produced change in practice. The cases in which documented AI bias led to meaningful remediation were cases in which deploying organizations faced reputational, regulatory, or legal pressure. Buolamwini and Gebru's (2018) Gender Shades audit prompted public commitments from

named companies to improve their systems. Regulatory frameworks that formalize and strengthen that accountability, rather than treating it as optional, are therefore the most promising path toward systemic improvement.

5. Addressing Counterarguments

5.1 The Programmer Accountability Objection

The strongest version of the programmer accountability argument holds that individual technical professionals often know more about a system's failure modes than their employers, and that this knowledge creates personal moral obligations. This is true, and it does not conflict with the primary corporate responsibility argument. Companies can and should be held primarily responsible for the organizational decision to deploy a system; individual programmers can additionally be held accountable for gross negligence or deliberate concealment of known risks. These are not competing frameworks but complementary ones operating at different levels of analysis.

5.2 The Historical Causation Objection

The argument that companies cannot be held responsible for bias rooted in social history they did not create is philosophically serious but practically insufficient as a basis for non-accountability. The relevant question is not whether a company created historical bias, but whether it chose to use data that encodes that bias in a consequential automated system without adequate safeguards. A pharmaceutical company is not responsible for creating a disease, but it is responsible for ensuring its drugs are safe before putting them on the market. The analogy is apt: companies that deploy AI systems in high stakes domains are responsible for the known and foreseeable failure modes of those systems, regardless of the historical origin of the bias.

5.3 The Technical Impossibility Objection

A third objection holds that because bias cannot be fully eliminated from systems trained on historical data, holding companies accountable for its presence is unreasonable. This objection mis frames what accountability requires. Accountability does not demand perfection; it demands due diligence. The standard should not be whether a system is entirely free of bias no system is but whether the deploying organization took reasonable and systematic steps to identify known failure modes, disclose them to affected parties, implement available mitigations, and establish mechanisms for ongoing monitoring and correction.

The technical impossibility objection also conflates the impossibility of eliminating all bias with the much weaker claim that nothing can be done to reduce it. Raji et al. (2020) demonstrate that systematic auditing processes can identify and significantly reduce disparate impact in many cases. The claim that accountability is futile because perfection is impossible does not survive contact with the empirical evidence on what auditing and accountability mechanisms can achieve.

6. Conclusions and Research Gaps

This survey paper has examined the empirical literature on AI bias, analyzed the three candidate loci of responsibility, and situated the analysis within existing and proposed regulatory frameworks in the EU and Saudi Arabia. The central conclusion is that deploying companies bear primary ethical and legal responsibility for algorithmic harm, grounded in their control over the deployment decision, their capacity to audit and remediate, their financial benefit from the system's outputs, and the organizational nature of the accountability gap identified in the literature.

This conclusion carries practical implications. It suggests that effective accountability for AI bias requires regulatory requirements focused primarily on deploying organizations: mandatory pre deployment bias auditing for high risk applications, mandatory disclosure of training data characteristics and known performance disparities, individual rights to explanation and human review for consequential automated decisions, and clear liability rules that create financial incentives for organizations to invest in bias prevention rather than treating it as a post hoc compliance issue.

The framework does not eliminate programmer accountability. Professional ethics standards for AI practitioners comparable to the standards that govern engineers, physicians, and lawyers would complement organizational accountability without replacing it. A data scientist who knowingly conceals a model's discriminatory behavior from their employer and from affected users has acted unethically regardless of what their employment contract says, and professional accountability mechanisms should reflect that.

Several important research gaps remain. First, there is limited empirical research on AI bias from the Gulf and Arab regional context; most existing literature reflects U.S. and European institutional settings, and it is not safe to assume that findings transfer directly across contexts with different historical patterns of discrimination and different institutional structures. Second, the relationship between SDAIA's principles framework and operational enforcement mechanisms has not been examined in published research. Third, comparative legal analysis of how different jurisdictions are

operationalizing AI accountability remains underdeveloped. Fourth, the literature on programmer professional responsibility in AI contexts is less developed than the corporate accountability literature. Fifth, longitudinal studies tracking whether regulatory interventions actually reduce disparate impact in deployed systems are absent. Addressing these gaps will be important as AI deployment scales in Saudi Arabia and the broader region, and as policymakers seek evidence-based guidance for regulatory design.

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